

Yves Alain Iragena

GRADUATE RESEARCH ASSISTANT | M.S. DATA ANALYTICS

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PROFESSIONAL SUMMARY

Systems engineer and Data Analytics researcher specializing in **LLMs, multi-agent architectures, and zero-latency business observability tools**. Proven track record of designing and deploying highly scalable, production-ready AI frameworks combining retrieval-augmented generation and multimodal machine learning to solve enterprise-level challenges.

TECHNICAL EXPERTISE

Architecture:	Autonomous AI Agents, Retrieval-Augmented Generation (RAG), Enterprise API Design
Languages & Frameworks:	Python, PyTorch, FastAPI, LangGraph, JavaScript, Plotly.js
Infrastructure:	Vector Databases, SQL, Docker, GitHub Actions, AWS

PROFESSIONAL EXPERIENCE

Multimodal AI Laboratory (MAIL) 2025 – Present

Graduate Research Assistant

- Architected defense mechanisms for AI integrity, focusing on signal processing and multimodal model safety.
- Collaborated with faculty to translate theoretical ML research into deployable detection frameworks.

Health Target Parapharmacy Ltd 2023 – 2024

Data & Operations Analyst | Kigali, Rwanda

- Automated 20+ hours of weekly manual data processing via Python and SQL engineering pipelines.
- Integrated automated inventory tracking logic, systematically reducing data entry errors by 70%.

FLAGSHIP ENGINEERING PROJECTS

AuraCommerce | BI & Observability ↗

- Designed an enterprise command center executing real-time data streaming and metrics polling autonomously without layout thrashing.
- High-Performance **FastAPI** Master Node executing telemetry processing under 50ms latency.
- Built "Glassmorphism" frontend integrating **Plotly.js** logic.

Auto Agent SEO Engine (JP AutoHub) ↗

- LangGraph**-based multi-agent system executing automated content workflows to optimize technical retail SEO.
- Connected intelligent agent pipelines directly to Vector Databases for high-speed semantic memory retrieval.

Deepfake Audio Detection Model ↗

- Hybrid CNN and Vision Transformer classification pipeline targeting synthetic voice cloning artifacts.
- Achieved a competitive **0.94 F1-score** matching state-of-the-art benchmarks in multimodal research.

Contextual Gmail AI Agent ↗

- Autonomous pipeline integrating **Python, MCP, and LangChain**.
- Analyses and categorizes enterprise Gmail environments dynamically via automated tool-calling behavior loops.

Agentic RAG Security Sandbox ↗

- Engineered defensive architectures to intercept adversarial prompt injections within LLM orchestration layers.
- Benchmarked multiple vector-database retrieval topologies, successfully reducing semantic hallucinations by 82%.

Distributed Log Anomaly Pipeline ↗

- Scalable **PyTorch** and data analytics architecture processing high-frequency enterprise backend signals.
- Implemented time-series forecasting to rapidly detect distributed system latency deviations with 97% precision.

EDUCATION

Master of Science in Data Analytics

Present

The Catholic University of America — Washington, D.C.

- Recipient: Graduate Research Assistantship in Multimodal AI Systems.
- GPA: 3.7 | Focus on Advanced AI Infrastructure.

Bachelor of Science in Computer Science

Earned

Koneru Lakshmaiah University — Vijayawada, India

- Honors & Distinction | CGPA: 8.9/10